



State Machine & Per-State Control

GreenPinkCamera Injectable Teardown

Flexiv Rizon4 · 6-state FSM · primitive / Cartesian / joint-impedance control

The Controller Has Two Layers

Describe every state with BOTH layers

Layer 1 - Flexiv RDK Control Mode

NRT_PRIMITIVE_EXECUTION

Runs Flexiv primitives: MovePTP, MoveL, MoveC, InsertComp, FloatingCartesian, Contact, GraspComp, ZeroFTSensor

NRT_CARTESIAN_MOTION_FORCE

Direct operational-space motion / force (grasp descent)

NRT_JOINT_IMPEDANCE

Joint hold: SendJointPosition + SetJointImpedance (park arm rigidly during gripper-only actions)

Layer 2 - Control Paradigm

the Force / Joint / Operational language

JOINT SPACE

MovePTP (joint space, % vel scale), joint-impedance hold, the cap twist

OPERATIONAL / CARTESIAN

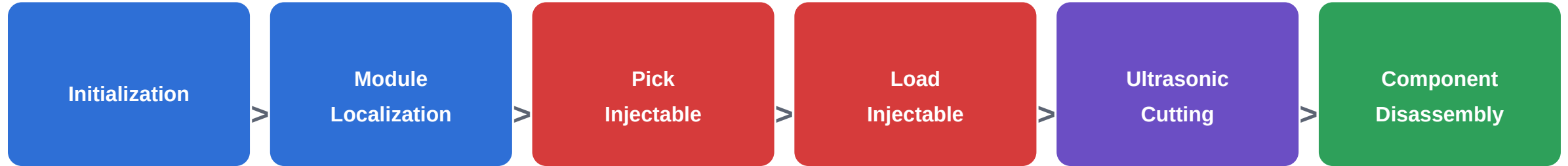
MoveL, MoveC - commanded in m/s

FORCE / COMPLIANCE

InsertComp, FloatingCartesian, Contact, GraspComp, Cartesian impedance, gripper 80 N, vise 5 kg

The State Machine

6 states · transition guards make it a real FSM



Transition guards (from recipe safety gates)

> enter vise area	Arduino busy=false, faulted=false, blade_on=false, homed
> cut	vise CLOSED, robot clear, rotary ~ 0
> after cut	CUT_HEIGHT=DONE, blade_on=false, x=z=0
fault edge (any state)	> STOP_ALL (fault handler)

Controller Legend

Color code reused on every state slide



JOINT

NRT_JOINT_IMPEDANCE

Primitives:

MovePTP · joint hold · cap twist



OPERATIONAL

NRT_CARTESIAN_MOTION_FORCE

Primitives:

MoveL · MoveC (m/s)



FORCE / COMPLIANCE

force-aware primitives

Primitives:

InsertComp · FloatingCartesian · Contact · GraspComp · gripper 80 N · vise 5 kg

State 1 - Initialization

Bring robot, gripper and machine to a known safe state

CONTROL:

Joint-space

Gripper

Primitive / action sequence

1. gripper open

2. MovePTP > Inter

3. MovePTP > Home

RDK control mode

NRT_PRIMITIVE_EXECUTION

Defining parameters

$\text{MOVE_JNT_VEL_SCALE} = 80\% \cdot \text{gripper } 0.05 \text{ m/s}$

State 2 - Module Localization

Re-localize the vise via ChArUco before any vise interaction

CONTROL:

Joint-space

Vision

Primitive / action sequence

1. MovePTP > Vise-cali

2. ChArUco detect (cali tag 1/2/3)

3. reload Vise.json

4. MovePTP > Home

RDK control mode

NRT_PRIMITIVE_EXECUTION

Defining parameters

hand-eye camera_tcp.yaml · tag_01_to_vise_tcp.json

State 3 - Pick Injectable

Camera-aligned approach + contact-guided adaptive grasp

CONTROL:

Operational

Force (contact grasp)

Primitive / action sequence

1. MovePTP > Plate

2. Align Injectable (MoveL)

3. ZeroFTSensor > Contact > GraspComp

4. MoveL lift +20 cm

RDK control mode

**PRIMITIVE_EXECUTION +
CARTESIAN_MOTION_FORCE**

Defining parameters

align 0.02 m/s · contact 0.05 m/s · grip 80 N

State 4 - Load Injectable

SHOWCASE STATE - cycles through all three RDK modes

CONTROL:

Force/compliance

Joint hold

Operational

Primitive / action sequence

1. MovePTP > above_vise

2. InsertComp (-Z > TCP X)

3. FloatingCartesian y/z/rx while vise > 5 kg

4. joint hold

5. gripper release

6. MoveL retreat

RDK control mode

ALL THREE RDK MODES

Defining parameters

INSERTCOMP_INSERT_VEL = 0.02 ·
maxContactForce · float maxVel 0.2 m/s

State 5 - Ultrasonic Cutting

Machine cuts while robot waits clear; then grip + twist off cap

CONTROL:

Machine

Force-float

Joint twist

Primitive / action sequence

1. Arduino CUT_HEIGHT (robot idle/clear)

2. MoveL down

3. FloatingCartesian y/z/rx

4. gripper 80 N

5. twist about TCP X 0>+7>-10 deg

6. MoveL lift

RDK control mode

NRT_PRIMITIVE_EXECUTION

Defining parameters

CUT z=134.2, x=110.5, deg=360 · twist scale 20%

State 6 - Component Disassembly

Fast-mode pick-and-place of spring, plastic, shell & glass

CONTROL:

Joint transit

Operational

Force-grip

Primitive / action sequence

1. MovePTP transit (xN)

2. MoveL descend / lift 0.5 m/s

3. gripper 80 N

4. Arduino OPEN_VISE

5. MoveC dump arc

RDK control mode

NRT_PRIMITIVE_EXECUTION

Defining parameters

FRAME5_CARTESIAN_VEL = 0.5 m/s · dump 0.03 m/s

Two Things To Remember

1 . Color-code by control paradigm

Blue = joint, green = operational, red = force - consistent across the FSM and every state slide. Pick & Load read as force-rich; Disassembly is mostly kinematic.

2 . Load Injectable is the showcase state

It is the only state that cycles through all three RDK modes: compliant InsertComp > floating force > joint-impedance hold > Cartesian retreat. This is your 'we understand control architecture' slide.